EoS Prediction Widget: Predictions 3, 4 and the Cubic Approximation (5.1 / 5.2)

Notation

A network $f : \mathbb{R}^p \to \mathbb{R}^M$ ($M = N \cdot d_{\text{out}}$ for N samples), parameters $\theta \in \mathbb{R}^p$, per-output $f_k(\theta)$. Jacobian $J \in \mathbb{R}^{M\times p}$ with row k equal to ∇f_k; residual $r = y - f \in \mathbb{R}^M$ (MSE); per-sample Hessian $Q_k = \nabla^2 f_k(\theta) \in \mathbb{R}^{p\times p}$; third-derivative tensor $T_k = \nabla^3 f_k(\theta)$ with the double contraction $T_k[a, b] = \nabla^3 f_k \cdot [a, b] \in \mathbb{R}^p$. We write the NTK $= JJ^\top$, the *residual-contracted gradient* $J^\top r = \sum_k r_k \nabla f_k$, and the *residual-weighted function Hessian* $M_r = \sum_k r_k Q_k$. Gradient descent uses $\Delta\theta = -\,\text{scale} \cdot g$ with $g = J^\top r$ and scale $= \frac{\eta}{N}$ for plain GD (a preconditioner P and step scale generalise this; $g = P\, J^\top r$).

Two curvature contractions recur throughout and are the crux of Predictions 3 vs. 4:

$$\textbf{QJr}\ \text{(exact, per sample)}: \quad \mathrm{d}J_k \;=\; \tfrac{\eta}{N}\, Q_k\big[J^\top r\big] \;=\; \tfrac{\eta}{N}\, Q_k\Big(\textstyle\sum_j r_j \nabla f_j\Big), \tag{1}$$

$$\textbf{rQJ}\ \text{(approximation)}: \quad \mathrm{d}J_k \;=\; \tfrac{\eta}{N}\, M_r\, \nabla f_k \;=\; \tfrac{\eta}{N}\Big(\textstyle\sum_a r_a Q_a\Big)\nabla f_k. \tag{2}$$

QJr applies the k-*th* Hessian to the *residual-contracted gradient*; rQJ applies the *residual-weighted Hessian sum* to the k-*th gradient*. They coincide only in special cases; the widget uses QJr in Prediction-3 and the (deliberate) rQJ approximation in Prediction-4.

1 Prediction-3 — frozen-J linear residual theory

1.1 Anchor (new): the Phase-1 threshold

The theory starts at the first iteration $t^\star$ at which the **Phase-1 alignment** reaches a threshold τ (hyperparameter `p3thr`, default 0.8):

$$\boxed{\frac{\big|(J^\top r)^\top u_1\big|}{\|J^\top r\|} \;\geq\; \tau} \qquad u_1 = \text{top (most positive) eigenvector of } M_r = \textstyle\sum_k r_k Q_k. \tag{3}$$

u_1 is obtained by a matrix-free Lanczos eigendecomposition of M_r (via the HVP $v \mapsto M_r v$). This replaces the old trigger (the first sign change of $d = \|J\dot{r}\| - \|\dot{J}r\|$). At $t^\star$ we freeze $J^\star = J(t^\star)$, $r^\star = r(t^\star)$, $\theta^\star = \theta(t^\star)$. All theories may *re-anchor* to the live run every `p3rep` iterations.

1.2 The three residual theories (each advanced one GD step per actual step)

(1) First order — frozen-J linear NTK. J held at $J^\star$:

$$r_{s+1} = \Big(I - \tfrac{\eta}{N}\, J^\star J^{\star\top}\Big) r_s, \qquad r_0 = r^\star \quad \Big(\text{general: } r_{s+1} = r_s - \text{scale}\, J^\star\big(P J^{\star\top} r_s\big)\Big). \tag{4}$$

(2) Second order — frozen $J^\star$ and $Q^\star$. With $g_2 = J^{\star\top} r_2$,

$$r_{2,\,s+1} = r_{2,s} \;-\; \text{scale}\left(J^\star g_2\right) \;-\; \tfrac{1}{2}\,\text{scale}^2\left(g_2^\top Q_k^\star g_2\right)_{k=1}^{M}, \tag{5}$$

i.e. the quadratic Taylor of $\Delta f_k = J_k^\star \cdot \Delta\theta + \frac{1}{2}\Delta\theta^\top Q_k^\star \Delta\theta$ enters the residual with a minus sign ($r = y - f$).

(3) Evolving-J — true QJr power iteration (step function). A "window-J" $\hat{J}$ is held fixed for $k = $ `p3k` steps while the residual decays through it; a *shadow* $\hat{J}^{\text{adv}}$ advances *every* step by the exact QJr recurrence; every k steps the window jumps to the shadow:

$$\begin{aligned}
r_{s+1} &= \left(I - \tfrac{\eta}{N}\,\hat{J}\hat{J}^\top\right) r_s, && \text{(r decays via the constant window-J)} && (6)\\
\hat{\theta}_{s+1} &= \hat{\theta}_s + \text{scale}\left(\hat{J}^\top r_s\right), && \text{(self-trajectory; where Q is read)} && (7)\\
\hat{J}_k^{\text{adv}} &\mathrel{+}= \tfrac{\eta}{N}\,Q_k(\hat{\theta})\left[\hat{J}^{\text{adv}\top} r\right], && \textbf{QJr, every step} && (8)\\
\hat{J} &\leftarrow \hat{J}^{\text{adv}} && \text{every k steps.} && (9)
\end{aligned}$$

Yes, Q changes: Q_k is re-evaluated at the self-trajectory $\hat{\theta}$ (not frozen). The Jacobian used in $r_{s+1} = (I - \frac{\eta}{N}\hat{J}\hat{J}^\top) r_s$ therefore comes from the **QJr** recurrence (line 4), *not* rQJ.

1.3 Plots

Left: the diagnostic $d = \|J\dot{r}\| - \|\dot{J}r\|$ (its two norms and difference). *Middle:* $\|r\|$ actual vs. the three theories (log). *Right:* $\cos\left(r_{\text{actual}}, r_{\text{theory}}\right)$. The *multi-freeze* strip repeats theory (1)/(2) with $J^\star$ frozen at $t^\star{-}10$, $t^\star{+}10$, $t^\star{+}20$ (these offsets track the new Phase-1 anchor automatically).

2 Prediction-4 — frozen-M_r Jacobian propagation (rQJ)

At t_0 freeze θ_0, $J_0 = J(\theta_0)$, r_0. Both variants advance J by the **rQJ** / $M_r \cdot J$ recurrence

$$\boxed{J_{s+1} = \left(I + \tfrac{\eta}{N}\,M_r\right) J_s} \quad\Longleftrightarrow\quad \Delta J_k = \tfrac{\eta}{N}\,M_r\,\nabla f_k, \qquad M_r = \textstyle\sum_a r_a Q_a, \tag{10}$$

then report the predicted NTK spectrum $\lambda\left(J_s J_s^\top\right)$ (top-3) vs. the actual NTK eigenvalues. This is *deliberately* the rQJ approximation of the true $\mathrm{d}\hat{J}_k = \frac{\eta}{N}\,Q_k[\hat{J}^\top \hat{r}]$ (QJr) that Prediction-3 uses — the panel exists to show how well rQJ tracks the real NTK.

Variant A — frozen residual. $M_r = \sum_a r_{0,a}\,Q_a(\theta_0)$ is *fully frozen* at t_0 (Q at θ_0, residual r_0).

Variant B — evolving-Qr. $\hat{J}_{s+1} = \left(I + \text{scale}\,M_r(\hat{r}_Q, \hat{\theta}_Q)\right)\hat{J}_s$ with $M_r = \sum_a \hat{r}_{Q,a}\,Q_a(\hat{\theta}_Q)$ **refreshed every $s = $ `p4s` steps** (held constant within a window $\Rightarrow$ a true $(I + \text{scale}\,M_r)^s$ power iteration). Here $\hat{\theta}_Q$ is the GD self-trajectory and $\hat{r}_Q$ the *bounded quadratic self-residual*

$$\hat{r} = Y - \left(f_0 + J_0\,\Delta\hat{\theta} + \tfrac{1}{2}\,\Delta\hat{\theta}^\top Q_0\,\Delta\hat{\theta}\right), \qquad \Delta\hat{\theta}_{s+1} = \Delta\hat{\theta}_s + \text{scale}\,g. \tag{11}$$

So Variant B *does* re-evaluate Q (at $\hat{\theta}_Q$, per s-window) and evolve r, but the contraction is still rQJ ($M_r \cdot J$), not QJr. Both variants re-anchor to the live run every `p4rep` iterations.

3 Predictions 5.1 (Trace) and 5.2 (Sharpness) — cubic approximation

Both panels forecast an NTK statistic from a frozen local model, re-anchored to the actual state every window. Four trajectories: {quadratic, cubic} $\times$ {live, self residual}. At each anchor freeze θ_0, J_0, f_0, and (cubic only) the third-derivative tensor $T = \nabla^3 f(\theta_0)$.

3.1 Shared Jacobian propagation

Per step set $g = J^\top r$, scale $= \frac{\eta}{N}$. Then

$$\begin{aligned} \textbf{Quadratic}\ (T=0,\ Q\ \text{frozen}) : \Delta J &= \text{scale}\, Q_0[g], && (12) \\ \textbf{Cubic}\ (Q\ \text{propagated},\ T \neq 0) : \Delta J &= \text{scale}\, Q_t[g] + \tfrac{1}{2}\,\text{scale}^2\, T[g,g], && (13) \\ Q_t[g] &= Q_0[g] + \sum_{g_k \in \text{hist}} \text{scale}\, T[g_k, g]. \end{aligned}$$

$Q_0[g] = \big(\nabla^2 f_k(\theta_0)\, g\big)_k$ is a per-sample HVP; $T[a,b]$ is a central difference of that HVP, $T[a,b] \approx \|a\|\big(Q(\theta_0 + \varepsilon\hat{a})[b] - Q(\theta_0 - \varepsilon\hat{a})[b]\big)/2\varepsilon$ ($\varepsilon = 10^{-2}$). The residual is either **live** $r = r(\theta_t)$ (measured from the run) or **self** $r = Y - \big(f_0 + J_0\Delta\theta + \frac{1}{2}\Delta\theta^\top Q_0 \Delta\theta\big)$ with $\Delta\theta_{s+1} = \Delta\theta_s + \text{scale}\, g$ (a fully closed forecast, no live residual).

3.2 5.1 — Trace of the NTK

Using $\|J + \Delta J\|_F^2 = \|J\|_F^2 + 2\langle J, \Delta J\rangle_F + \|\Delta J\|_F^2$, the panel accumulates, per step, the predicted $\mathrm{Tr}(\mathrm{NTK}) = \|J\|_F^2$ two ways:

$$\textbf{with PSD:}\quad \mathrm{Tr}(\mathrm{NTK})_s \,/\, N \;=\; \|J_s\|_F^2 \,/\, N, \tag{14}$$

$$\textbf{without PSD:}\quad \mathrm{Tr}(\mathrm{NTK})_s \,/\, N \;=\; \Big(\|J_0\|_F^2 + \sum 2\,\langle J, \Delta J\rangle_F\Big)\Big/ N \qquad (\text{drops } \|\Delta J\|_F^2 \geq 0). \tag{15}$$

Overlaid on the two *actual* edges (all $\div N$): $\mathrm{Tr}(J^\top J) = \|J\|_F^2$ (exact, Gauss–Newton) and $\mathrm{Tr}(\nabla^2 L)$ (full loss Hessian, Hutchinson with 24 Rademacher probes of the exact loss-Hessian HVP).

3.3 5.2 — Sharpness $\sigma_1 = \lambda_{\max}(\mathrm{NTK})$

Let u_1 be the top NTK eigenvector ($JJ^\top u_1 = \sigma_1 u_1$) and $v_1 = J_0^\top u_1 / \|J_0^\top u_1\|$ the paired top right-singular vector. Under $J \to J + \Delta J$ the top NTK eigenvalue perturbs as

$$\boxed{\Delta\sigma_1 \;=\; \underbrace{2\sqrt{\sigma_1}\, v_1^\top \Delta J^\top u_1}_{\text{first order}} \;+\; \underbrace{\big\|\Delta J^\top u_1\big\|^2}_{\text{PSD term}}\;.} \tag{16}$$

Summing the per-step contributions with the cubic ΔJ gives (accumulators `A51`, `P51`)

$$\sigma_1^{\text{pred}} \,/\, N = \Big(\underbrace{\lambda_{\max}(J_0 J_0^\top)}_{\text{Base}} \;+ A51\; \big[+\; P51\big]\Big)\Big/ N, \qquad P51 = \sum \big\|\Delta J^\top u_1\big\|^2, \tag{17}$$

$$A51 = \sum \Big[\; \underbrace{2\,\text{scale}\,\sqrt{\sigma_1}\; v_1^\top \big(Q_t[g]\big)^\top u_1}_{\text{quadratic part}} \;+\; \underbrace{\text{scale}^2\,\sqrt{\sigma_1}\; v_1^\top \big(T[g,g]\big)^\top u_1}_{\text{cubic } \eta^2 \text{ part}} \;\Big]. \tag{18}$$

The optional "cubic without η^2" curve drops only the cubic η^2 part from $A51$ (keeping the same cubic trajectory and PSD term). Overlaid on the actual edges $\lambda_{\max}(\nabla^2 L)$ (loss-Hessian sharpness) and $\lambda_{\max}(JJ^\top)$ (NTK). These are the multi-sample Eq.-51 forecasts; the single-sample analogue (Eq.-47) is

$$\Delta J = \eta r\, Q_t\, J + \tfrac{1}{2}\eta^2 r^2\, T[J, J], \qquad \Delta\sigma_1 = 2\, J^\top \Delta J + \|\Delta J\|^2. \tag{19}$$

All formulas mirror `server.py` (`cubic_step`, the `g_trace` / sharpness forecast, and the Prediction-3/4 blocks) and the fp64 `eos_lab` reference. `scale=` η/N for plain GD; a preconditioner replaces $g = J^\top r$ by $g = P\, J^\top r$ and η/N by the optimizer's step.